SZABIST
University

Shaheed Zulfikar Ali Bhutto
Institute of Science and Technology

ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

Total Marks:_____

Obtained Marks: _____

Computing Vision

Lab Task # 04

Submitted To: **Mr. Muhammad Azhar**

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Reg. Number: **23108109**

Contrast Stretching

Contrast stretching of a low contrast grayscale image using OpenCV and matplotlib.

A local fruit inspection company is digitizing their quality control process. They use overhead cameras to capture images of fruits like apples, oranges, and bananas on a conveyor belt. However, due to poor lighting conditions in the warehouse, the captured images often appear dull and low in contrast, making it difficult for both human inspectors and AI systems to analyze them properly.

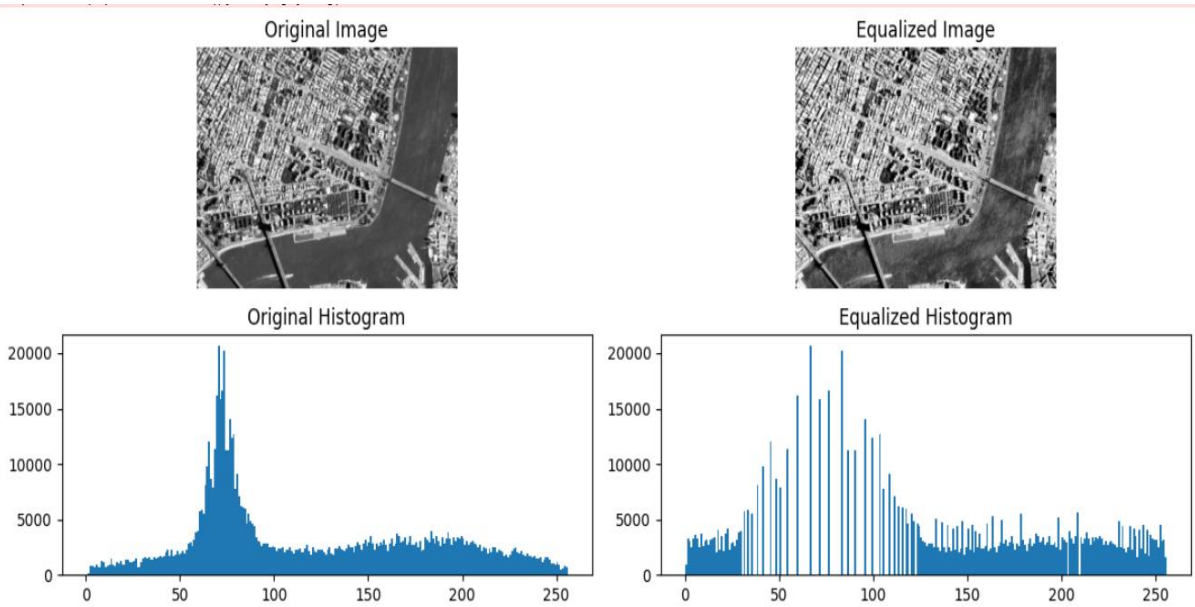
As a Python developer and computer vision enthusiast, you've been asked to improve the visibility and contrast of these grayscale fruit images using contrast stretching.

```
[1]: # Contrast Stretching
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("D:/23108109/FF2.jpg", cv2.IMREAD_GRAYSCALE)
min_val = np.min(img)
max_val = np.max(img)
stretched = ((img - min_val) / (max_val - min_val)) * 255
stretched = stretched.astype(np.uint8)
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.title("Original Image")
plt.imshow(img, cmap='gray')
plt.axis('off')
plt.subplot(1, 2, 2)
plt.title("Contrast Stretched")
plt.imshow(stretched, cmap='gray')
plt.axis('off')
plt.tight_layout()
plt.show()
cv2.imwrite("D:/23108106/AA1.jpg", stretched)
```

Histogram Equalization

A remote sensing team is working with grayscale satellite imagery collected during a cloudy day. Due to poor lighting conditions and haze, the satellite images suffer from low contrast, making it difficult to identify terrain features like roads, rivers, and boundaries. The team wants to enhance these grayscale images using a basic image processing technique — histogram equalization — to stretch the contrast and make the details more visible.

```
[2]: # Histogram Equalization
img = cv2.imread("D:/23108109/F7.jpg", cv2.IMREAD_GRAYSCALE)
equalized = cv2.equalizeHist(img)
plt.figure(figsize=(12, 5))
plt.subplot(2, 2, 1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis('off')
plt.subplot(2, 2, 2)
plt.imshow(equalized, cmap='gray')
plt.title("Equalized Image")
plt.axis('off')
plt.subplot(2, 2, 3)
plt.hist(img.ravel(), 256, [0,256])
plt.title("Original Histogram")
plt.subplot(2, 2, 4)
plt.hist(equalized.ravel(), 256, [0,256])
plt.title("Equalized Histogram")
plt.tight_layout()
plt.show()
```

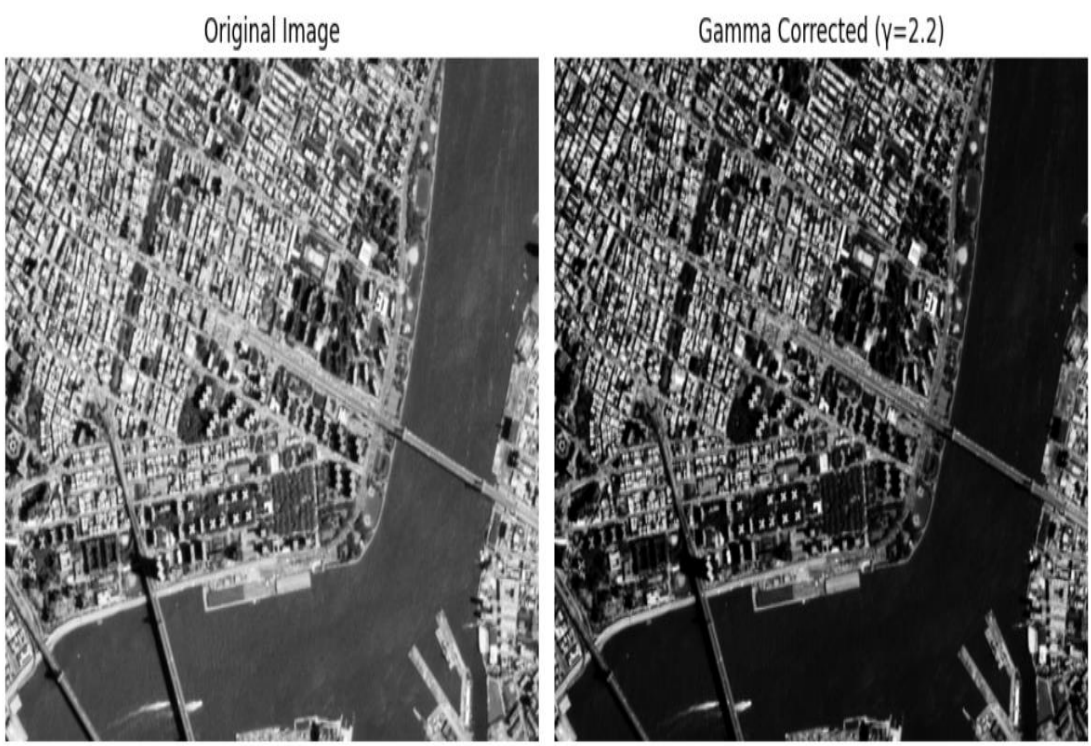


Gamma Correction:

A geospatial intelligence team is analyzing satellite images captured during early morning and late evening hours. These images are inherently low in contrast due to insufficient lighting and atmospheric scattering. This makes it difficult for analysts to distinguish critical features such as roads, vegetation, river boundaries, and urban zones.

While traditional histogram equalization may over-enhance certain areas, the team needs a more controlled enhancement method that preserves visual balance while improving feature visibility.

```
[3]: # Gamma Correction
img = cv2.imread("D:/23108109/F7.jpg")
img_gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
gamma = 2.2
gamma_corrected = np.array(255 * (img_gray / 255) ** gamma, dtype='uint8')
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow(img_gray, cmap='gray')
plt.title("Original Image")
plt.axis('off')
plt.subplot(1, 2, 2)
plt.imshow(gamma_corrected, cmap='gray')
plt.title(f"Gamma Corrected (γ={gamma})")
plt.axis('off')
plt.tight_layout()
plt.show()
```



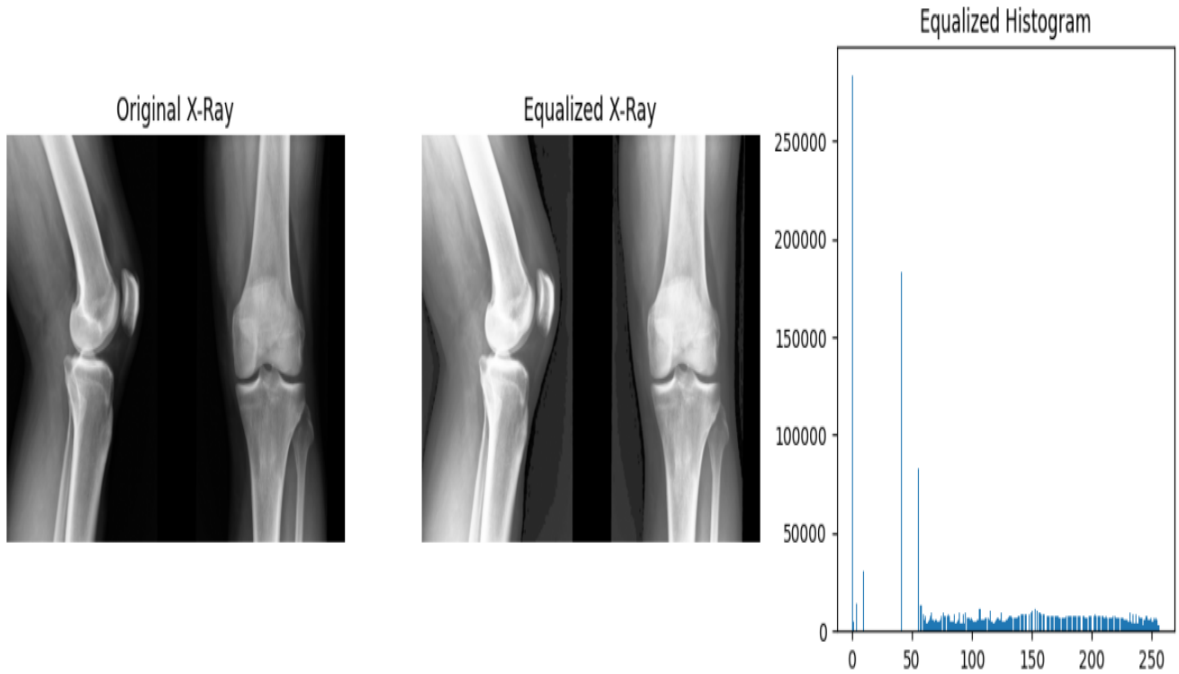
Case Study 1: Medical Imaging

A radiology department receives low-contrast chest and bone X-rays captured with suboptimal exposure. These images make it difficult to differentiate between bone, soft tissue, and possible pathological regions (e.g., fractures or lung lesions).

Objective: Enhance contrast using histogram equalization to help radiologists spot tissue boundaries and anomalies more clearly.

```
[4]: # Case Study 1: Medical Imaging
img = cv2.imread("D:/23108109/F8.jpg", cv2.IMREAD_GRAYSCALE)
equalized = cv2.equalizeHist(img)
plt.figure(figsize=(12,4))
plt.subplot(1,3,1)
plt.imshow(img, cmap='gray');
plt.title("Original X-Ray");
plt.axis('off')
plt.subplot(1,3,2)
plt.imshow(equalized, cmap='gray');
plt.title("Equalized X-Ray");
plt.axis('off')
plt.subplot(1,3,3)
plt.hist(equalized.ravel(),256,[0,256])
plt.title("Equalized Histogram")
plt.tight_layout()
plt.show()
```

```
C:\Users\dell\AppData\Local\Temp\ipykernel_16208\1740564176.py:14: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is d
eprecated since Matplotlib 3.10; the parameter will become keyword-only in 3.12.
  plt.hist(equalized.ravel(),256,[0,256])
```



Case Study 2: Satellite Image Enhancement

A remote sensing team is analyzing satellite images taken under hazy or overcast conditions. The images have poor contrast, obscuring features such as water bodies, vegetation, roads, and farmland boundaries.

Objective: Apply gamma correction and histogram equalization to enhance visibility of terrain and land cover details.

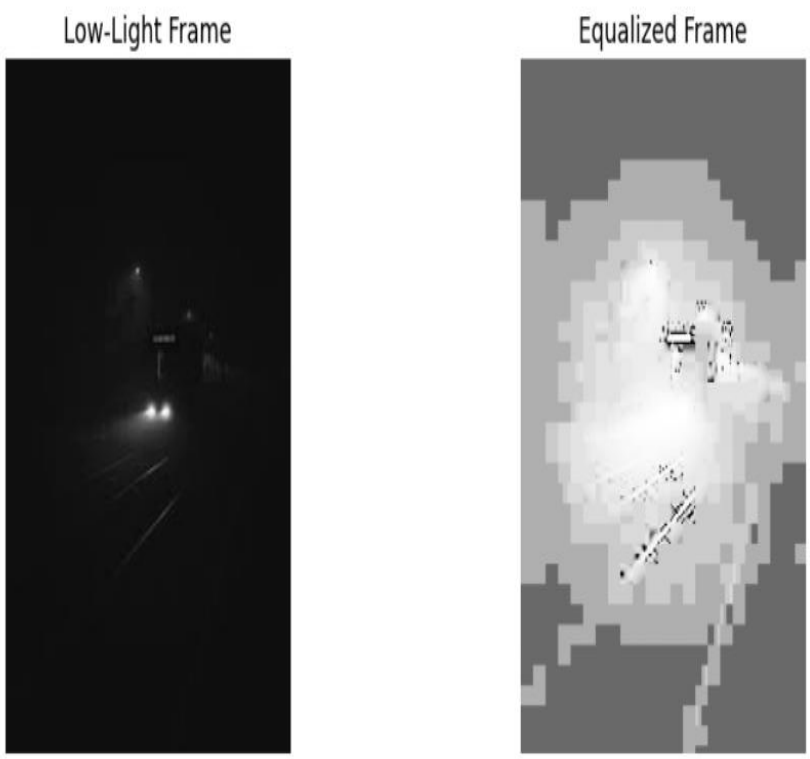
```
[5]: # Case Study 2: Satellite Image Enhancement
img = cv2.imread("D:/23108109/F7.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
equalized = cv2.equalizeHist(gray)
gamma = 0.5
gamma_corr = np.array(255 * (gray/255) ** gamma, dtype='uint8')
plt.figure(figsize=(15,5))
plt.subplot(1,3,1)
plt.imshow(gray, cmap='gray');
plt.title("Original Gray");
plt.axis('off')
plt.subplot(1,3,2)
plt.imshow(equalized, cmap='gray');
plt.title("Histogram Equalized");
plt.axis('off')
plt.subplot(1,3,3)
plt.imshow(gamma_corr, cmap='gray');
plt.title(f"Gamma Corrected (γ={gamma})");
plt.axis('off')
plt.tight_layout()
plt.show()
```

Case Study 3: Surveillance Video Processing

A security surveillance system captured nighttime video footage where facial details and features are obscured by low light. These frames make it hard for facial recognition software (and human operators) to identify subjects.

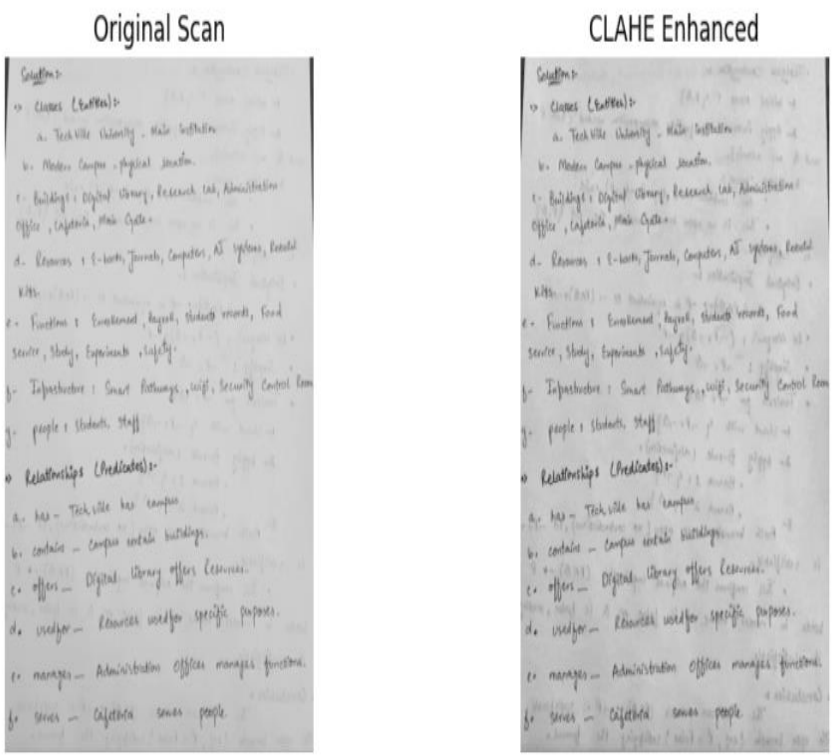
Objective: Use histogram equalization to enhance contrast in grayscale video frames, making facial features and edges more distinct.

[6]:
```
# Case Study 3: Surveillance Video Processing
img = cv2.imread("D:/23108109/F9.jpg", cv2.IMREAD_GRAYSCALE)
equalized = cv2.equalizeHist(img)
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plt.imshow(img, cmap='gray');
plt.title("Low-Light Frame");
plt.axis('off')
plt.subplot(1,2,2)
plt.imshow(equalized, cmap='gray');
plt.title("Equalized Frame");
plt.axis('off')
plt.tight_layout()
plt.show()
```



Case Study 4: Document Image Enhancement (OCR Preparation)
An archival digitization project involves scanning hundreds of old, low-quality paper documents. These scanned documents have poor contrast, faded ink, and uneven lighting, which reduces the accuracy of OCR (Optical Character Recognition) systems.
Objective: Use adaptive histogram equalization (CLAHE) to enhance contrast locally and prepare the images for better OCR results.

[7]:
```
# Case Study 4: Document Image Enhancement (OCR Preparation)
img = cv2.imread("D:/23108109/FF5.jpeg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
enhanced = clahe.apply(gray)
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap='gray');
plt.title("Original Scan");
plt.axis('off')
plt.subplot(1, 2, 2)
plt.imshow(enhanced, cmap='gray');
plt.title("CLAHE Enhanced");
plt.axis('off')
plt.tight_layout()
plt.show()
```

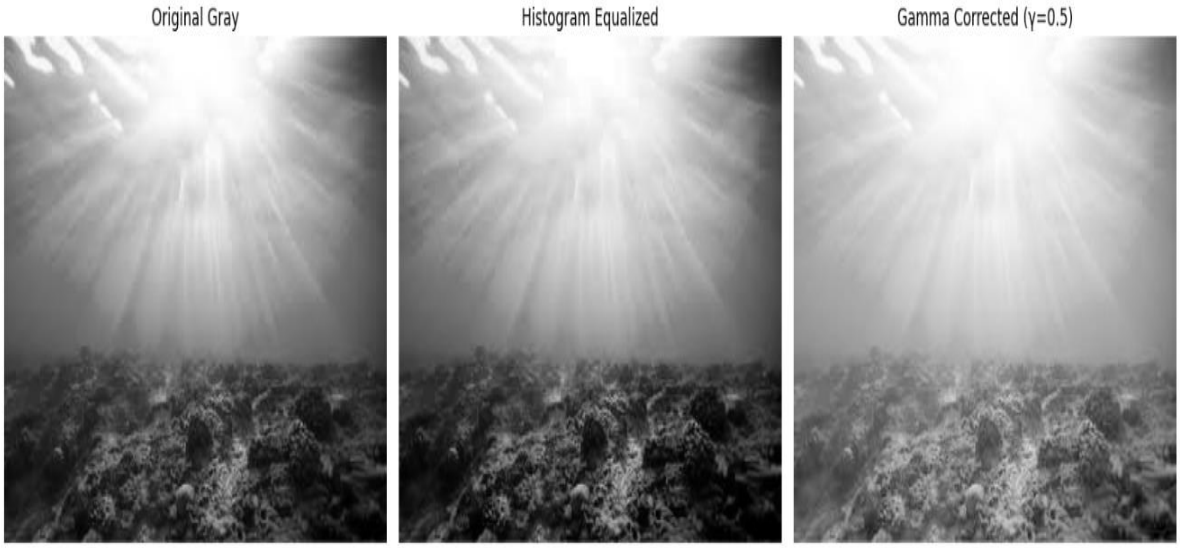


Case Study 5: Foggy Road Image Enhancement for Driver Assistance
An automotive AI system receives input from a dashcam driving in foggy or rainy conditions. Road signs, lane markings, and pedestrians are barely visible due to low visibility and reduced contrast.
Objective: Apply contrast stretching (min-max normalization) to improve visibility of objects in foggy images for better perception and driver assistance.

```
[8]: # Case Study 5: Foggy Road Image Enhancement for Driver Assistance
img = cv2.imread("D:/23108109/F10.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
min_val = np.min(gray)
max_val = np.max(gray)
stretched = ((gray - min_val) / (max_val - min_val) * 255).astype('uint8')
gamma = 0.4
gamma_corrected = np.array(255 * (stretched / 255) ** gamma, dtype='uint8')
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.imshow(gray, cmap='gray');
plt.title("Original Foggy");
plt.axis('off')
plt.subplot(1, 3, 2)
plt.imshow(stretched, cmap='gray');
plt.title("Contrast Stretched");
plt.axis('off')
plt.subplot(1, 3, 3)
plt.imshow(gamma_corrected, cmap='gray');
plt.title(f"Gamma Corrected (γ={gamma})");
plt.axis('off')
plt.tight_layout()
plt.show()
```

Case Study 1: Underwater Image Enhancement for Marine Biology
Scenario:
Underwater photos suffer from poor visibility and low contrast due to water scattering and absorption. Marine biologists struggle to identify coral health and marine life in these images.
Objective: Use histogram equalization and gamma correction to enhance contrast and improve feature visibility in underwater scenes.

```
[9]: # Case Study 1: Underwater Image Enhancement for Marine Biology
img = cv2.imread("D:/23108109/F11.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
equalized = cv2.equalizeHist(gray)
gamma = 0.5
gamma_corr = np.array(255 * (gray/255) ** gamma, dtype='uint8')
plt.figure(figsize=(15,5))
plt.subplot(1,3,1)
plt.imshow(gray, cmap='gray');
plt.title("Original Gray");
plt.axis('off')
plt.subplot(1,3,2)
plt.imshow(equalized, cmap='gray');
plt.title("Histogram Equalized");
plt.axis('off')
plt.subplot(1,3,3)
plt.imshow(gamma_corr, cmap='gray');
plt.title(f"Gamma Corrected (γ={gamma})");
plt.axis('off')
plt.tight_layout()
plt.show()
```



Case Study 2: Enhancing Nighttime Road Scenes for Autonomous Vehicles
Scenario:
Autonomous vehicles rely on cameras that often capture noisy and low-contrast images at night, making lane detection and obstacle avoidance challenging.
Objective: Apply adaptive histogram equalization (CLAHE) to selectively enhance local contrast without amplifying noise, aiding in safer navigation.

```
[10]: # Case Study 2: Enhancing Nighttime Road Scenes for Autonomous Vehicles
img = cv2.imread("D:/23108109/F9.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
enhanced = clahe.apply(gray)
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap='gray');
plt.title("Original Scan");
plt.axis('off')
plt.subplot(1, 2, 2)
plt.imshow(enhanced, cmap='gray');
plt.title("CLAHE Enhanced");
plt.axis('off')
plt.tight_layout()
plt.show()
```

Case Study 3: Enhancing Security Camera Footage in Poor Lighting
Scenario:
Security cameras installed in dimly lit areas produce dark, low-contrast video frames, reducing the effectiveness of surveillance and face recognition.
Objective: Use contrast stretching combined with gamma correction to brighten and clarify these frames in real-time processing.

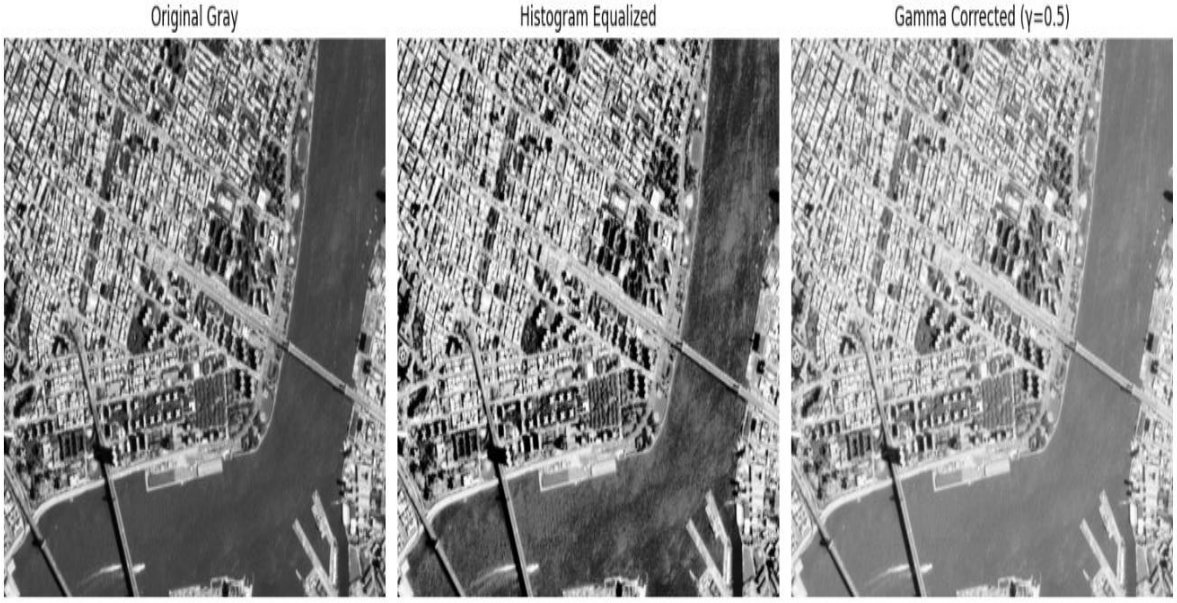
```
[11]: # Case Study 3: Enhancing Security Camera Footage in Poor Lighting
img = cv2.imread("D:/23108109/F9.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
min_val = np.min(gray)
max_val = np.max(gray)
stretched = ((gray - min_val) / (max_val - min_val) * 255).astype('uint8')
gamma = 0.4
gamma_corrected = np.array(255 * (stretched / 255) ** gamma, dtype='uint8')
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.imshow(gray, cmap='gray');
plt.title("Original Foggy");
plt.axis('off')
plt.subplot(1, 3, 2)
plt.imshow(stretched, cmap='gray');
plt.title("Contrast Stretched");
plt.axis('off')
plt.subplot(1, 3, 3)
plt.imshow(gamma_corrected, cmap='gray');
plt.title(f"Gamma Corrected (γ={gamma})");
plt.axis('off')
plt.tight_layout()
plt.show()
```

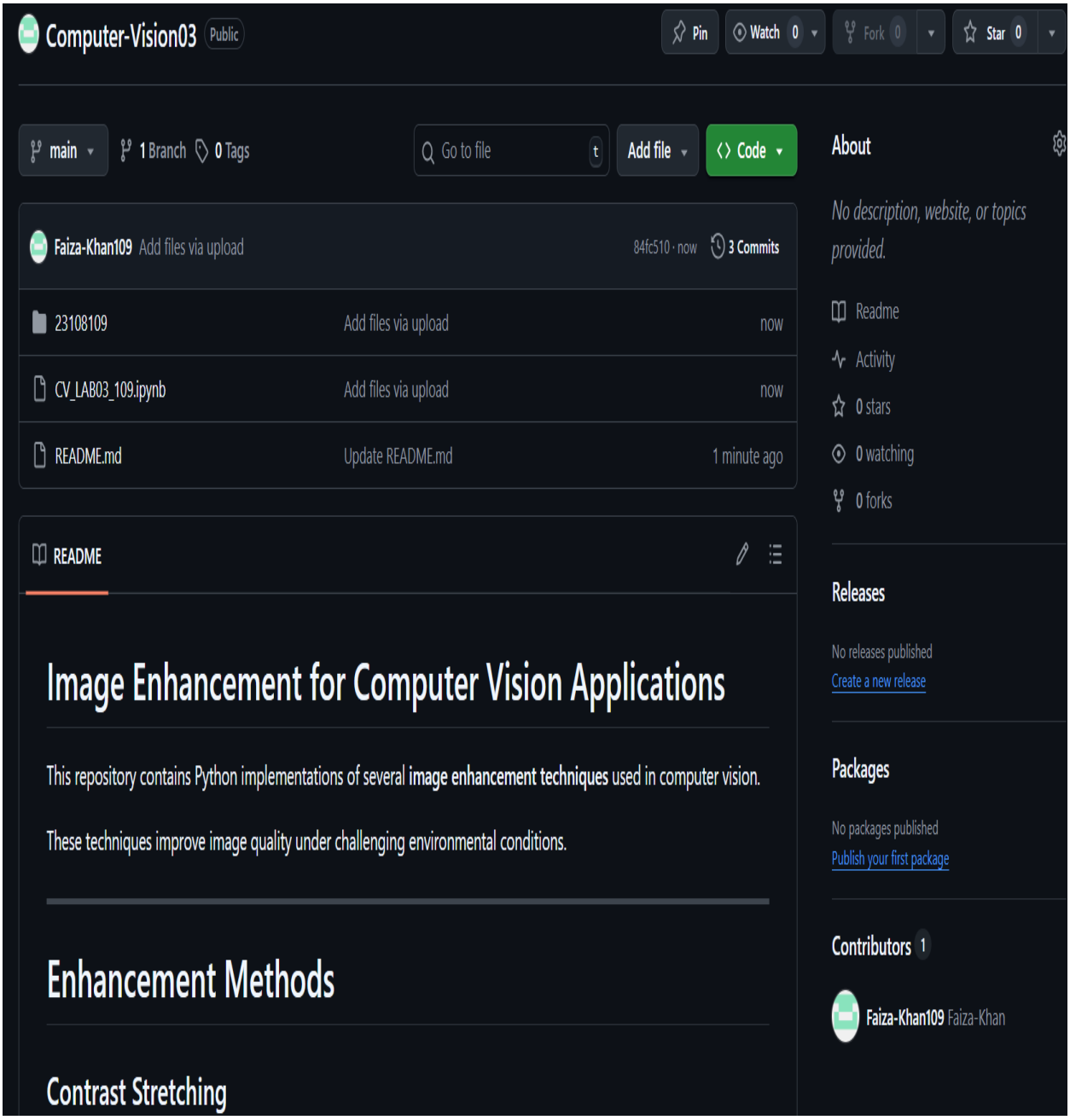
Case Study 4: Improving Visibility in Foggy or Smoggy Urban Environments
Scenario:
Traffic monitoring cameras face visibility issues due to pollution-induced haze, reducing contrast in captured images.
Objective: Enhance scene contrast via contrast stretching and histogram equalization to clarify vehicles, traffic signals, and pedestrians.

```
[12]: # Case Study 4: Improving Visibility in Foggy or Smoggy Urban Environments
img = cv2.imread("D:/23108109/F10.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
min_val = np.min(gray)
max_val = np.max(gray)
stretched = ((gray - min_val) / (max_val - min_val) * 255).astype('uint8')
equalized = cv2.equalizeHist(gray)
plt.figure(figsize=(12,4))
plt.subplot(1,3,1)
plt.imshow(gray, cmap='gray')
plt.title("Original Foggy Image")
plt.axis('off')
plt.subplot(1,3,2)
plt.imshow(stretched, cmap='gray')
plt.title("Contrast Stretched")
plt.axis('off')
plt.subplot(1,3,3)
plt.imshow(equalized, cmap='gray')
plt.title("Histogram Equalized")
plt.axis('off')
plt.tight_layout()
plt.show()
```

Case Study 5: Enhancing Satellite Thermal Images for Environmental Monitoring
Scenario:
Thermal satellite images often show low contrast, making it difficult to distinguish temperature gradients critical for forest fire detection or urban heat mapping.
Objective: Use gamma correction and histogram equalization to increase thermal contrast, improving feature detection.

[13]:
```
# Case Study 5: Enhancing Satellite Thermal Images for Environmental Monitoring
img = cv2.imread("D:/23108109/F7.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
equalized = cv2.equalizeHist(gray)
gamma = 0.5
gamma_corr = np.array(255 * (gray/255) ** gamma, dtype='uint8')
plt.figure(figsize=(15,5))
plt.subplot(1,3,1)
plt.imshow(gray, cmap='gray');
plt.title("Original Gray");
plt.axis('off')
plt.subplot(1,3,2)
plt.imshow(equalized, cmap='gray');
plt.title("Histogram Equalized");
plt.axis('off')
plt.subplot(1,3,3)
plt.imshow(gamma_corr, cmap='gray');
plt.title(f"Gamma Corrected (γ={gamma})");
plt.axis('off')
plt.tight_layout()
plt.show()
```





https://github.com/Faiza-Khan109/